EduPulse SMT Legal Counsel Dossier

Purpose

This is the single-file consolidated legal dossier for **EduPulse SMT**. It is intended to give counsel one forwardable document covering:

1. Business framing
2. Whole-system technical operation
3. Payment categorization, reconciliation, and historical route-metadata handling
4. Cryptography and AWS security posture
5. Intellectual-property and confidential-know-how positioning
6. Ghana data-protection technical controls
7. Founder-side legal action priorities
8. Draft internal legal templates for assignment and confidentiality

This document is a practical legal-support dossier, not legal advice.

Business Name

EduPulse SMT

SMT means **Smart School Management Technology**.

Counsel Cover Note

Subject: EduPulse SMT Legal Review Dossier for Incorporation, IP, and Data Protection

Hello Counsel,

I write to share a consolidated legal review dossier for **EduPulse SMT** to support work on:

1. Business incorporation
2. Intellectual-property ownership and protection
3. Confidentiality and contributor controls
4. Ghana data-protection positioning
5. Platform terms and school-facing legal documents

EduPulse SMT is a multi-tenant school SaaS platform with several technical and operational features that I want reflected accurately in the legal work:

1. Paystack-linked payment categorization, reconciliation, and legacy route-metadata handling
2. Application-layer encryption of selected sensitive data
3. Tamper-evident finance ledgering using cryptographic chaining
4. Role-gated academic, finance, and approval workflows
5. School-by-school tenant separation and access controls

The principal legal outcomes I need from this review are:

1. Proper legal framing of EduPulse SMT as a business and software asset
2. Strong ownership and confidentiality protection for the code, workflows, and operating materials
3. Clear data-protection positioning under Ghana law
4. Contract language that protects the platform while making school data responsibilities clear
5. Execution-ready contributor assignment and confidentiality forms

I would be grateful if you would pay particular attention to:

1. Which parts of the platform should be treated as confidential know-how or trade-secret-like assets
2. Whether the school contracts and terms should distinguish clearly between EduPulse SMT as platform operator and each school as controller of school-originated personal data
3. Whether stronger employee, contractor, founder, or partner paperwork is needed to secure IP ownership and confidentiality
4. Whether the draft assignment and confidentiality templates in this dossier should be revised into final execution-ready forms

Best regards, Daniel Asare Anyinam EduPulse SMT danielasareanyinam2020@gmail.com

Executive Brief

What EduPulse SMT Is

EduPulse SMT is a multi-tenant SaaS platform for schools in Ghana. It combines:

1. School operations
2. Student and academic workflows
3. Payment and billing workflows
4. Proprietor oversight and governance
5. Audit and security controls

It should be understood legally as more than a generic school app. It is an operating system for school workflow control, payment governance, and audit visibility.

Why This Matters Legally

Counsel should treat the platform as containing at least four legally important layers:

1. Proprietary workflow design
2. Proprietary payment-categorization and reconciliation logic
3. Confidential security and control architecture
4. Regulated personal-data handling for schools, parents, guardians, staff, and billing contacts

The Core Proprietary Value

The strongest proprietary elements appear to be:

1. Multi-tenant school-control architecture
2. Role-gated workflow and approval logic
3. Reference-first payment categorization and reconciliation logic
4. Reconciliation and recovery logic for missing payment writes
5. Tamper-evident ledgering for finance records

6. Combined use of encryption, audit logging, and routing controls inside one school operations platform

Paystack Positioning

EduPulse SMT uses Paystack in two main ways:

1. For school subscription and plan payments
2. For school fee collection using stable learner identity plus structured payment references

The payment-classification design is especially important. The active model centers on structured references, school-controlled fee-category rules, and reconciliation logic so incoming payments can still be classified, audited, and recovered consistently. Legacy route metadata may still appear in older records and remains relevant for historical audit interpretation.

Cryptography Positioning

EduPulse SMT uses cryptography in two distinct ways:

1. Application-layer encryption for sensitive stored values such as parent contacts and saved bank details
2. HMAC-based tamper-evident chaining for finance ledger entries

In the documented AWS production posture, the software is also designed to use managed secrets and encrypted infrastructure controls, including AWS Secrets Manager and encrypted backup/storage recommendations.

Ghana Data-Protection Positioning

The system processes personal data relating to students, guardians, staff, school operators, and billing records. Its documented controls include:

1. Role-based access control
2. Tenant isolation
3. Encryption in transit
4. Application-layer encryption of selected sensitive fields
5. Audit logging and incident response planning
6. Retention and deletion policy structure

Whole-System Technical Summary

Platform Model

EduPulse SMT is a multi-tenant SaaS platform for schools in Ghana. Each school operates as a logically separated tenant inside one platform. The system combines:

1. Academic operations
2. Payment and fee oversight
3. Subscription billing and school-plan management
4. Proprietor visibility and governance controls
5. Audit logging and tamper-evident ledgering

Main Roles

The platform recognizes roles including:

1. Proprietor
2. Headmaster
3. Teacher
4. Finance
5. Admin

These roles matter legally because the software encodes separation-of-duty rules. Certain financial actions are intentionally separated between recording and approval to reduce fraud risk and strengthen audit defensibility.

End-To-End Operational Flow

At a high level, the system operates as follows:

1. A school registers and selects a plan
2. If the plan is paid, the system initializes Paystack payment
3. Once payment is verified, the school subscription is activated
4. The proprietor can submit school verification details that are screened by an external KYB provider
5. Clean KYB outcomes can auto-verify; flagged outcomes are routed to review-required status for platform compliance decisions
6. School users log in under role-based access controls
7. Staff carry out academic and finance workflows
8. Certain finance events create ledger entries that are cryptographically chained
9. Sensitive data is encrypted at the application layer before being stored
10. In production, the documented AWS deployment posture injects secrets through managed secret storage and recommends encrypted storage and backups

Paystack Architecture And Payment Categorization

Two Distinct Payment Contexts

EduPulse SMT uses Paystack in at least two important contexts:

1. Subscription and plan payments for schools themselves
2. Fee collection for school-specific payment categories such as tuition, feeding, uniform, transport, and similar categories

School Subscription Payments

When a school chooses a paid plan, the system initializes a Paystack transaction and redirects the school to Paystack checkout. After Paystack confirms success, the platform activates the school's subscription.

The system also keeps payment references and reconciliation logic so successful payments that fail to write locally can later be recovered.

School Fee Categorization By Structured Reference

For school operations, the active implementation uses a stable learner payment identity plus structured references. The operating idea is:

1. The platform generates a stable reference format tied to the school, learner, and intended fee category
2. Parents or payers use that reference when making payment
3. Paystack callback data returns the reference and transaction details
4. The system classifies the payment from the structured reference and school payment rules
5. Reconciliation logic can recover or correct records when a gateway success was not fully written locally

What “Split” Means In This System

In EduPulse SMT, the Paystack “split” idea survives mainly as historical routing-and-settlement metadata. The implementation concepts are:

1. Older school records can still include legacy route metadata by payment type
2. Those records may store route labels, provider markers, destination details, or other earlier routing identifiers from older payment flows
3. Historical payment initialization and webhook payloads may still carry route identifiers that remain relevant for audit interpretation
4. Related settlement or reconciliation metadata may therefore still appear in older ledger records

This means the software is not merely collecting one undifferentiated payment stream, but its currently supported categorization model is structured-reference based rather than active fee-route or split setup.

Webhook Security And Idempotency

Paystack webhooks are protected using HMAC SHA-512 signature verification against the raw request body and the Paystack secret key.

Before writing a Paystack finance payment, the system checks whether a payment with the same reference already exists. If so, it returns a duplicate outcome instead of recording the payment again.

Reconciliation And Recovery

EduPulse SMT includes reconciliation logic that can fetch successful Paystack transactions over a lookback window and recover transactions that succeeded at Paystack but were not fully recorded locally.

Cryptography And Security Architecture

Application-Layer Encryption

The platform defines an `EncryptedString` type backed by the Python `cryptography` library's `Fernet` implementation.

In production:

1. The application requires a `DATA_ENCRYPTION_KEY`
2. Startup is blocked if that key is missing
3. Values are encrypted before being written to the database

4. Values are decrypted when read back through the ORM

Sensitive fields shown in current models include:

1. Saved bank details
2. Certain school email fields
3. Guardian and parent names
4. Parent phone numbers
5. Parent email addresses

Tamper-Evident Financial Ledger

Separate from field encryption, the platform uses a cryptographic ledger chain for financial records:

1. Each confirmed finance event becomes a ledger entry
2. Each entry includes the previous hash
3. A new HMAC-SHA256 hash is computed over the entry payload plus the previous hash
4. The chain starts from a genesis value if no prior entry exists
5. If any stored finance record is altered later, the verification chain can reveal the tampering

The ledger uses a dedicated `LEDGER_SECRET_KEY`, and production startup is blocked if that key is missing.

AWS Production Posture

The documented production deployment describes an AWS-oriented model that uses:

1. ECS task definitions
2. Secret injection from AWS Secrets Manager
3. Encrypted storage recommendations for EBS-backed volumes
4. Encrypted backups to S3 using SSE-KMS

The legal meaning is that EduPulse SMT performs application-layer cryptography itself while AWS is the recommended environment for secret storage and encrypted infrastructure controls.

IP Asset Inventory

Asset Categories

The most relevant IP categories here are:

1. Copyright in source code, documentation, designs, and templates
2. Trade secrets and confidential know-how in system workflows and controls
3. Brand assets and naming
4. Contractual protection of methods, implementation detail, and operating playbooks

Core Asset Inventory

Asset	Description	Likely Protection Type	Why It Matters
Multi-tenant school	Tenant-specific school scoping, role controls, and	Trade secret, copyright, confidential know-how	Core system structure for serving many

Asset	Description	Likely Protection Type	Why It Matters
architecture	operational separation		schools in one platform
Role-gated workflow engine	Proprietor, headmaster, teacher, finance, admin workflow separation	Trade secret, confidential know-how	Encodes governance logic and internal-control design
Payment categorization architecture	Structured-reference payment classification, historical route compatibility, and reconciliation controls	Trade secret, confidential know-how, copyright in code	Differentiates fee collection, payment classification, and audit recovery
Reconciliation recovery logic	Recovery of successful gateway payments not fully recorded locally	Trade secret, confidential know-how	Protects revenue integrity and operational reliability
Tamper-evident finance ledger	HMAC-chained ledger entries and verification logic	Trade secret, confidential know-how, copyright in implementation	Strong compliance and audit differentiator
Application-layer encrypted fields	ORM-backed encrypted storage of sensitive values	Copyright in implementation, confidential security know-how	Protects personal and banking data
Subscription and metering logic	Plan selection, plan state, overage, trial, downgrade handling	Trade secret, copyright, confidential know-how	Commercial model embedded in code
Academic approval workflows	Report staging, approval, and governance loops	Trade secret, copyright, confidential know-how	Part of the control-layer value proposition
Operations documentation system	SOPs, launch packets, handoff docs, incident templates, onboarding playbooks	Copyright, confidential business know-how	Turns software into a repeatable operating business
Brand and product naming	EduPulse SMT name and presentation materials	Trademark/brand, copyright in materials	Customer and legal identity

High-Value Trade Secret Candidates

The strongest trade-secret candidates appear to be:

1. The school-by-school payment categorization controls and fee-type rules
2. The reference-validation and reconciliation rules used to classify and recover verified payments
3. The reconciliation sweep and missing-payment recovery design
4. The dual use of encrypted fields plus tamper-evident finance ledgering

5. The role-gated separation between recording, approval, audit review, and proprietor oversight
6. The full operations layer that connects onboarding, billing, launch, support, incidents, and renewal management

Ghana Data Protection Technical Brief

Scope Of Data Processed

The platform processes data relating to:

1. Students
2. Parents and guardians
3. School staff and operators
4. Proprietors and school administrators
5. Billing and payment records
6. Security, authentication, and audit events

Role Positioning: School vs Platform

The existing privacy documentation points to this distinction:

1. Each school is the primary controller of school-originated student and guardian data in its operational context
2. EduPulse SMT acts as the platform operator and processor/service provider for those schools in many processing activities

Technical Controls Currently Evident

1. Role-based access control
2. School/tenant isolation
3. Restricted access to sensitive finance and audit functions
4. TLS encryption in transit
5. Application-layer encryption at rest for selected sensitive fields
6. HMAC-SHA256 tamper-evident logging for finance records
7. HMAC SHA-512 verification of Paystack webhook events
8. External KYB provider integration with explicit verified/review-required outcomes and audit logging
9. Secrets management and production hardening
10. Incident and breach handling documentation
11. Retention, deletion, and DSAR workflow documentation

Practical Risk Areas To Review

1. Placeholder legal fields still awaiting formal company registration details or address information
2. Whether all high-risk data flows have matching contractual language
3. Whether school-side user behavior could still cause improper data entry or disclosure despite platform controls
4. Whether internal roles and named owners are formalized in corporate documents, not only operating docs

Founder IP And Confidentiality Checklist

Immediate Priorities

1. Confirm the legal entity name to be used for ownership of the platform
2. Confirm whether EduPulse SMT will be registered personally first or through a formal company entity immediately
3. Ensure all code created personally is clearly attributable to you or the business entity
4. Confirm there is no contributor code in the repo without written ownership or assignment terms
5. Put invention-assignment language in every developer or contractor agreement
6. Put confidentiality clauses in every contractor or collaborator agreement
7. Ensure contracts say schools own their school data and EduPulse SMT owns the platform, code, workflows, and materials
8. Preserve repository history, screenshots, architecture notes, and dated packet docs as evidence

Strongest Assets To Protect First

If time is limited, prioritize legal protection around:

1. Multi-tenant school-control architecture
2. Payment categorization, legacy-route compatibility, and reconciliation logic
3. Tamper-evident financial ledger design
4. Security and encrypted-data handling design
5. Operations playbooks and school rollout materials

Draft Template 1: Contributor IP Assignment

Purpose

This is a draft internal template for assigning intellectual-property rights in work created for EduPulse SMT.

Draft Form

Contributor Intellectual Property Assignment Agreement

This Contributor Intellectual Property Assignment Agreement is made on [Date] between:

1. **EduPulse SMT / [Registered Entity Name]**, of [Address] (the "Company")
2. **[Contributor Full Name]**, of [Address] (the "Contributor")

The Contributor irrevocably assigns to the Company, with full title guarantee, all present and future right, title, and interest in and to the Work Product and all associated Intellectual Property Rights, to the fullest extent permitted by law.

For this purpose, **Work Product** means all software code, source files, scripts, database models, APIs, technical designs, prompts, documents, templates, playbooks, specifications, business methods, visual materials, communications materials, and derivative works created, authored, modified, or contributed by the Contributor in connection with EduPulse SMT.

The Contributor also agrees:

1. To sign further documents reasonably needed to confirm ownership
2. To disclose any pre-existing materials before using them
3. To keep EduPulse SMT confidential information confidential
4. Not to assert moral rights against the Company to the extent permitted by law

Draft Template 2: Confidentiality And NDA

Purpose

This is a draft confidentiality and non-disclosure template for contributors, advisors, contractors, prospective partners, or other parties given access to EduPulse SMT confidential information.

Draft Form

Confidentiality and Non-Disclosure Agreement

This Confidentiality and Non-Disclosure Agreement is made on [Date] between:

1. **EduPulse SMT / [Registered Entity Name]**, of [Address] (the "Disclosing Party")
2. **[Recipient Full Name or Entity]**, of [Address] (the "Receiving Party")

Confidential Information includes all non-public information relating to EduPulse SMT, including source code, software architecture, product roadmap, payment-categorization and reconciliation logic, legacy route-account compatibility structures, cryptographic controls, operations playbooks, onboarding materials, incident templates, pricing and commercial strategy, technical know-how, and draft legal materials.

The Receiving Party agrees:

1. To use Confidential Information only for the permitted purpose
2. To keep it strictly confidential
3. Not to copy, disclose, publish, reverse engineer, or exploit it except as authorized in writing
4. To limit access to those who genuinely need to know and who are bound by equivalent confidentiality obligations
5. To return or destroy the information on request or at the end of the relationship, subject to legal retention obligations

Repository Evidence Points

The strongest evidence files supporting this dossier include:

1. `backend/routes.py`
2. `backend/api/models.py`
3. `backend/security/crypto.py`
4. `backend/paystack_integration.py`
5. `docs/security/AUDIT_INTEGRITY_SYSTEM.md`
6. `SECURITY.md`
7. `compliance/PRIVACY_POLICY.md`
8. `compliance/DATA_RETENTION_POLICY.md`
9. `compliance/INCIDENT_RESPONSE_PLAN.md`
10. `docs/operations/internal/STANDARD_OPERATING_PROCEDURES.md`

Recommended Counsel Questions

1. What is the best legal framing for EduPulse SMT as a business entity and software product?
2. Which system elements should be explicitly reserved as IP, confidential information, or trade-secret-like know-how?
3. What contributor, employee, contractor, and partner agreements are needed to ensure ownership remains with EduPulse SMT?
4. How should Ghana data-protection roles be described between EduPulse SMT and each school?
5. What terms should govern customer data ownership, software ownership, audit evidence, and misuse restrictions?
6. Should the included draft assignment and NDA templates be refined into final execution versions for immediate use?

Recommended Use

Use this dossier when you want one single document to forward to counsel without sending a multi-file packet first. For working edits and modular review, the separate files in `docs/operations/internal/` remain the canonical breakdown.